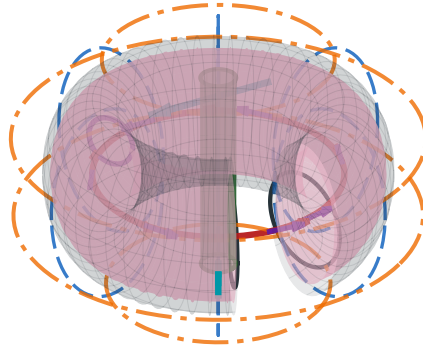


Atlas checkpoint — what has actually been derived so far

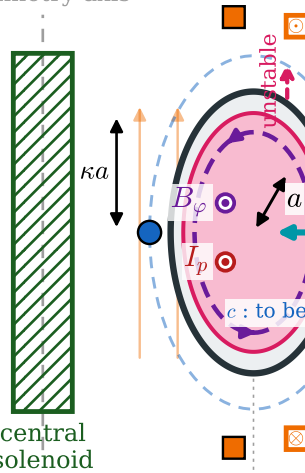
every component on the machine at this point, numbered by the section that forced it. Nothing later appears.

perspective view (vessel cut open)



- vacuum vessel (chamber wall)
- plasma / fuel region
- toroidal-field (TF) coils
- central solenoid (CS)
- poloidal-field (PF) coils
- magnetic field B
- plasma current I_p
- auxiliary heating ports

symmetry axis



poloidal cross-section (R - Z plane)

central solenoid

R_0 major radius

⊙ : the vector points out of the page

component → the section that forced it

- §4 vacuum vessel / torus the corridor is closed: R_0 and a exist
- §4 plasma boundary two lengths, hence $A = R_0/a$
- §4 TF coils, B_φ $B_\varphi \propto 1/R$ around the same tube
- toroidal drift the bottle leaks; a second field is forced
- §6 central solenoid, I_p B_θ comes from I_p , nothing else
- 7 pitch, safety factor q the helix has a pitch, and it is bounded
- §9 PF / shaping coils pressure pushes out; B_v holds position
- §12 NBI / RF port the transformer cannot reach T_{fusion}
- §13 elongation κ more room at fixed R_0
- §14 sensors + feedback a taller plasma tips: shaping needs a loop

NOT here, because not yet derived: triangularity, alpha heating, the neutron boundary, blanket, shield, superconductors, divertor, and every spherical-tokamak feature. A component enters this table only after a section has forced it.

Idealized schematic; not a fabrication drawing.

$R_0/a = 3.0$, $\kappa = 1.70$, $\delta = 0.00$, 6 of 12 TF coils in the 3-D panel; all geometry computed by tokamak_build.py.